

5. Git-HOL

Clean Up and Push Back to Remote Git

What is "Clean Up and Push Back to Remote Git"?

After you finish your work (such as adding files, resolving conflicts, or merging branches), you should:

1. Make sure your local repository is clean.
2. Remove unnecessary branches or files.
3. Push your latest changes to the remote repository (GitHub/GitLab).

This keeps both your local and remote repositories organized and up to date.

Step 1: Check Repository Status

Before pushing, verify that everything has been committed.

`git status`

Output

On branch master

nothing to commit, working tree clean

Meaning:

- All changes are committed.
 - No untracked or modified files remain.
 - The repository is clean.
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Step 2: View All Branches

Check which branches exist.

`git branch`

Example:

* master

GitWork

The * indicates the current branch.

Step 3: Delete Unnecessary Local Branches

If a branch has already been merged, delete it.

```
git branch -d GitWork
```

Output:

Deleted branch GitWork (was a8b5c3f).

If Git refuses because the branch is not merged:

```
git branch -D GitWork
```

-d safely deletes a merged branch, while -D force deletes it.

Step 4: Remove Unwanted Files

If temporary or backup files exist, add them to .gitignore.

Example .gitignore:

*.orig

*.log

*.tmp

Stage and commit the updated .gitignore:

```
git add .gitignore
```

```
git commit -m "Updated .gitignore"
```

Step 5: Verify Commit History

Check your commits before pushing.

```
git log --oneline
```

Example:

```
7e4d8a2 Updated .gitignore
```

```
4b6c123 Resolved merge conflict
```

```
2f5a781 Added hello.xml
```

Step 6: Check the Remote Repository

View the configured remote.

```
git remote -v
```

Example:

```
origin https://github.com/username/MyProject.git (fetch)
```

```
origin https://github.com/username/MyProject.git (push)
```

Step 7: Push Changes to Remote

Push your local commits to GitHub or GitLab.

```
git push origin master
```

If your main branch is named main, use:

```
git push origin main
```

Example output:

```
Enumerating objects: 10
```

```
Counting objects: 100%
```

```
Writing objects: 100%
```

```
To https://github.com/username/MyProject.git
```

Now the remote repository contains your latest changes.

Step 8: Verify Everything is Clean

Run:

```
git status
```

Output:

On branch master

Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

This confirms:

- Local and remote repositories are synchronized.
- No pending changes remain.

Common Commands

Command	Purpose
git status	Check repository status
git branch	List local branches
git branch -d branch_name	Delete a merged branch
git branch -D branch_name	Force delete a branch
git remote -v	Display remote repository URLs
git log --oneline	Show commit history
git push origin master	Push changes to the remote repository
git push origin main	Push changes if the default branch is main

Example Workflow

git status

git branch

git branch -d GitWork

git add .gitignore

git commit -m "Updated .gitignore"

git log --oneline

git remote -v

git push origin master

git status